

 OR-AS Operations Research Applications and Solutions	Case Name: Social Housing	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	Jens Martens		User defined distributions
Date	2019	Project Control	Automatic tracking
File Name	C2019-02 Social Housing		Tracking based on user input

1. Project description

Project authenticity

The construction of social housing, including structural works, installations, and finishing activities.

The project consists of activity, resource, and cost data obtained directly from the actual project execution.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	19	Serial/Parallel (SP)	41%
Planned Duration (PD)	352 d	Activity Distribution (AD)	48%
Budget At Completion (BAC)	734,602.11€	Length of Arcs (LA)	4%
Renewable Resources	-	Topological Float (TF)	36%
Consumable Resources	-		

* standard eight-hour working days=

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	15	19	3.09
CRI-rho	21	21	2.00
CRI-tau	21	32	2.02

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	6	24	4.24
SI	9	8	0.94
SSI	6	24	4.24
CRI-r	14	23	3.50
CRI-rho	14	23	3.56
CRI-tau	13	23	3.70

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	N/A	N/A	1	N/A	N/A
PV - SPI	N/A	N/A	CPI	N/A	N/A
PV - SCI	N/A	N/A	SPI	N/A	N/A
ED - 1	N/A	N/A	SPI(t)	N/A	N/A
ED - SPI	N/A	N/A	SCI	N/A	N/A
ED - SCI	N/A	N/A	SCI(t)	N/A	N/A
ES - 1	N/A	N/A	0.8 CPI + 0.2 SPI	N/A	N/A
ES - SPI(t)	N/A	N/A	0.8 CPI + 0.2 SPI(t)	N/A	N/A
ES - SCI(t)	N/A	N/A			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 10 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	N/A	N/A	N/A	N/A	N/A	N/A	N/A
std dev	N/A	N/A	N/A	N/A	N/A	N/A	N/A
final	N/A	N/A	N/A	N/A	N/A	N/A	N/A

2.3.3.2. Time forecasting

PD	352	Real Duration	355	Late	0.2%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	N/A	N/A	N/A	N/A
PV - SPI	N/A	N/A	N/A	N/A
PV - SCI	N/A	N/A	N/A	N/A
ED - 1	N/A	N/A	N/A	N/A
ED - SPI	N/A	N/A	N/A	N/A
ED - SCI	N/A	N/A	N/A	N/A
ES - 1	N/A	N/A	N/A	N/A
ES - SPI(t)	N/A	N/A	N/A	N/A
ES - SCI(t)	N/A	N/A	N/A	N/A

2.3.3.3. Cost forecasting

BAC	734,602.11€	Real Cost	748,555.80€	Over budget	1.8%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	N/A	N/A	N/A	N/A
CPI	N/A	N/A	N/A	N/A
SPI	N/A	N/A	N/A	N/A
SPI(t)	N/A	N/A	N/A	N/A
SCI	N/A	N/A	N/A	N/A
SCI(t)	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI(t)	N/A	N/A	N/A	N/A